Dynamic Semantic Graph-Guided Cross-Modal Attention Network for Open-Vocabulary Object Detection in Aerial Images

Lingjun Li, Xinxin Zhang, Xiaoxu Feng, Yanbu Guo, Shigang Liu, Yali Peng, and Xiwen Yao, *Member, IEEE*

***Abstract*—Open-vocabulary object detection (OVD) in aerial images is critical for intelligent perception of multicategory objects in complex scenarios, yet still faces two key challenges: 1) the prevalence of long-tailed categories in aerial image domains leads to insufficient visual-semantic correlation modeling for novel classes due to annotation scarcity and 2) complex backgrounds induce feature confusion and degrade the robustness of cross-modal semantic alignment. To address these issues, we propose a dynamic semantic graph-guided cross-modal attention (DynaGraph-CrossAtt) Network for aerial open-vocabulary detection. The overall architecture of DynaGraph-CrossAtt is collaboratively constituted by two core components, i.e., a dynamic semantic-aware graph (DSG) convolutional module and a bidirectional cross-modal attention (BiCMA) mechanism. In particular, DSG constructs text semantic graphs to propagate fine-grained semantic relationships (e.g., part interactions and spatial dependencies) and dynamically updates graph node representations, thereby enhancing the contextual awareness and discriminability of text embedding, effectively alleviating the representation deficiency of novel classes. Subsequently, BiCMA receives the enhanced text semantic embedding output by DSG and performs cross-modal interaction with vision features. In BiCMA, the vision-to-text branch employs dynamic weight allocation to emphasize critical semantic nodes, while the text-to-vision branch leverages semantic graphs to guide vision feature focusing on object regions, suppressing background noise, forming a dynamic soft-alignment paradigm of "semantic-guided localization and vision feedback optimization" for cross-modal features. Experiments on aerial benchmarks demonstrate that DynaGraph-CrossAtt significantly improves OVD performance in complex aerial image scenarios.**

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Lingjun Li, Xinxin Zhang, and Yanbu Guo are with the School of Software, Zhengzhou University of Light Industry, Zhengzhou 450000, China.

Xiaoxu Feng is with the Institute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei 230026, China.

Shigang Liu and Yali Peng are with the School of Artificial Intelligence and Computer Science, Shaanxi Normal University, Xi'an 710021, China.

Xiwen Yao is with the School of Automation, Northwestern Polytechnical University, Xi'an 710021, China (e-mail: yaoxiwen@nwpu.edu.cn).

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***Index Terms*—Aerial images, cross-modal attention, graph convolution, open-vocabulary object detection (OVD).**

I. Introduction

OPEN-VOCABULARY object detection (OVD), as a cutting-edge research direction in computer vision, aims to transcend the constraints of traditional detection models that rely on predefined categories [1], [2], [3], [4]. It seeks to recognize and localize object classes not encountered during the training stage (i.e., novel classes) within images through semantic information. The technical essence of OVD lies in establishing a generalizable visual-semantic alignment mapping, enabling models to comprehend "unseen object concepts" via text, rather than depending on a fixed category label taxonomy. This paradigm shift holds significant importance for aerial image analysis, particularly demonstrating unique advantages and application prospects in complex scenarios such as drone inspection, aerial object surveillance, and disaster emergency response [5], [6], [7], [8], [9], [10], [11], [12].

Existing approaches to OVD predominantly leverage pretrained vision-language models (VLMs) like CLIP [13] and Grounding DINO [14] to construct cross-modal detection frameworks, broadly falling into two technical paradigms. The first paradigm generates class embedding via the text encoder and performs similarity matching with the vision feature [15]. For instance, YOLO-World [16] dynamically aligns CLIP-derived class embedding with vision features while utilizing an offline vocabulary for efficient inference; OWL-ViT [3] incorporates image-level patch embedding within Vision Transformers to align text and vision representations via contrastive loss and supports cross-modal retrieval using image or text prompts. The second paradigm transfers semantic knowledge from language models to detection networks via knowledge distillation, achieving novel class generalization through coarse-grained semantic alignment [17], [18]. Examples include HierKD [19], which bridges the VLM-detection model capability gap with hierarchical distillation losses; and Detic [20], which leverages image-level supervision to generate pseudobounding boxes and combines multimodal text embedding to improve detection accuracy for rare classes.

Although these methods achieve certain performance in general scenarios, their design schemes fundamentally misalign

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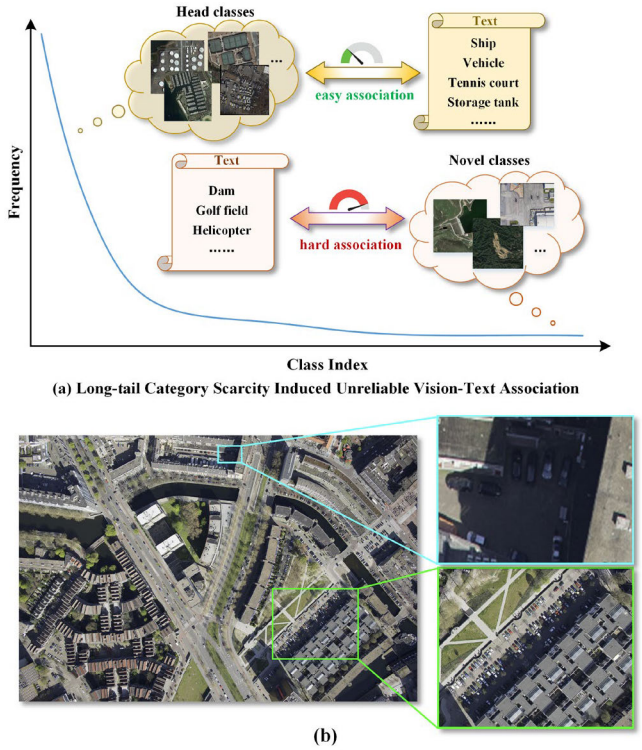


Fig. 1. (a) Semantic association and (b) background interference challenges of the OVD framework in aerial image scenes.

with the core demands of open-vocabulary aerial image object detection. Prevailing methods predominantly employ static semantic mapping strategies, relying solely on global semantic similarity for cross-modal matching (e.g., class-level alignment in YOLO-World), which struggles to capture the fine-grained semantic associations unique to aerial image objects (e.g., spatial constraint relationships). While knowledge distillation techniques are capable of transferring semantic knowledge, they exhibit excessive dependency on teacher model performance and lack dynamic adjustment mechanisms for cross-modal fusion, thus limiting their applicability to aerial image domain challenges. More critically, current research neglects two core aerial image-specific obstacles.

1) Long-tailed category scarcity hinders a reliable association between unknown-class vision patterns and text descriptions, resulting in poorly discriminative embedding for novel classes and further exacerbating semantic ambiguity, as shown in Fig. 1(a).
2) Overhead perspectives induce high background-object similarity, and densely arranged objects trigger vision feature confusion, resulting in cross-modal alignment drift, as shown in Fig. 1(b). These limitations collectively reveal that the current OVD frameworks lack systematic modeling capacity for the complex semantic correlations and background noise interference patterns inherent to aerial image scenarios.

To this end, we propose a dynamic semantic graph-guided cross-modal attention (DynaGraph-CrossAtt) network to enhance OVD performance in aerial images. DynaGraph-CrossAtt mainly consists of a dynamic semantic-aware (DSG) graph convolutional module and a bidirectional cross-modal attention (BiCMA) mechanism. In particular, first, DSG realizes the quantitative modeling of semantic relations among text nodes by constructing a text semantic graph, dynamically aggregating contextual semantic information using an adaptive graph convolution network, and iteratively updating the graph node representations, which effectively enhances the semantic generalization ability in the open-vocabulary scenarios and improves highly discriminative embedding representations of the novel classes. Subsequently, BiCMA receives the enhanced text semantic representation from DSG and performs cross-modal interaction with vision features. BiCMA mainly consists of two branches, one is the vision-to-text branch, which dynamically adjusts the weight of text embedding by using vision features, focuses on semantic nodes that are strongly correlated with vision features, and suppresses semantic drift under open vocabulary; the other is the text-to-vision branch, which utilizes the text semantic graph to guide the vision attention, and achieves background noise suppression through object region saliency enhancement. The two branches form a bimodal co-optimization framework of "semantic-guided localization and vision feedback optimization," which achieves a significant improvement in the robustness of cross-modal feature fusion under complex backgrounds.

The main contributions of this article as follows.

1) We propose a dynamic semantic-aware DSG, which effectively alleviates the problem of under-representation of novel classes in aerial images through a fine-grained relationship propagation and dynamic update mechanism for the text semantic graph.
2) We design a BiCMA mechanism to establish a semantic-guided vision feature focusing and a semantic weight adjustment mechanism for vision feedback, which significantly improves the anti-interference ability in complex backgrounds.
3) The effectiveness of the method is validated on several aerial image benchmark datasets, providing an innovative solution for open-vocabulary aerial image object detection.

II. Related Work

OVD in the field of remote sensing has achieved significant progress in recent years, with its core goal being to break the constraints of traditional closed-set detection and achieve zero-shot or few-shot detection of unseen classes during training [21], [22], [23], [24], [25], [26], [27]. Early research primarily relied on the generalization ability of cross-modal pretrained models (e.g., CLIP [13] and ALIGN [28]) to build a semantic space through visual–text alignment. For example, OVR-CNN [29] utilizes image–text data to extend the detection vocabulary and combines the detection data with image-level labels to improve the detection of unseen classes. Although such methods perform well in natural scenes, the high resolution, cluttered backgrounds, and diverse scales of objects in remote sensing images impose higher requirements on the model [30], [31], [32], [33]. Recent research in open-vocabulary object

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detection for remote sensing images has focused on data engineering, model innovation, and cross-modal fusion.

Data engines and the construction of large-scale datasets have emerged as a critical breakthrough driving advancement. Pan et al. [34] propose the LAE-DINO model, which achieves significant improvement in open-set detection performance on the LAE-1M dataset through dynamic vocabulary construction (DVC) and vision-guided text prompt learning, surpassing existing methods such as GLIP [35]. SegEarth-OV [36] effectively addresses the boundary blurring issue caused by low-resolution features by combining the training-free upsampler SimFeatUp [37] with global bias subtraction. GSNet [38] integrates remote sensing domain priors with the multimodal capabilities of CLIP, and significantly outperforms traditional methods in the open-vocabulary semantic segmentation task through dual-stream encoder and query-guided feature fusion [39]. OVA-DETR [40] employs region-text contrastive loss to replace traditional regression losses, enhancing small object detection efficiency and inference speed. Huang et al. [41] construct a 470 000-image dataset ORSD+ and propose OpenRSD, which supports text/image dual cue inputs, contains a multitasking detection header that jointly optimizes horizontal and directional box detection, and enhances cross-domain generalizability through multistage training, and the average accuracy exceeds that of YOLO-World in seven datasets, balancing efficiency and accuracy.

Cross-modal generation and data augmentation emerge as another key research direction. AeroGen [42] supports conditional generation using both horizontal and rotated bounding boxes. Synthetic data generated by AeroGen significantly improves the performance of detection models on datasets like DIOR [43]. TransOSS [44] constructs the first optical–SAR cross-modal vessel re-identification dataset, HOSS ReID, and achieves cross-modal feature alignment using a Transformer architecture. Furthermore, LAE-Label Engine [34] utilizes SAM [45] and a Large VLM (LVLM) for semiautomated annotation, constructing the LAE-1M dataset covering over 100 classes, providing a crucial data foundation for open-vocabulary detection.

Furthermore, **model efficiency and generalization capabilities** continue to be optimized as a key focus for researchers. MobileMamba [46] achieves a balance between speed and accuracy through multireceptive field feature interaction and lightweight design. RandAR [47] breaks the limitation of fixed-order generation, enabling zero-shot generalization for image tokens generated at arbitrary positions.

While current research in open-vocabulary detection has made progress in data scale, modality fusion, and generative augmentation, bottlenecks persist in modeling semantic relationships and noise robustness. To address this, our proposed DynaGraph-CrossAtt achieves structured dynamic evolution of class relationships and soft-alignment optimization of cross-modal features through its dynamic semantic graph and bidirectional attention mechanisms. Future work will further explore the self-evolution mechanism of semantic graphs and their lightweight deployment on edge devices.

III. Methodology

A. Overview

In this work, we aim to ameliorate the two major problems in open-vocabulary aerial image object detection, i.e., difficulties in modeling semantic associations due to scarce annotations for long-tailed categories and feature confusion coupled with fragile cross-modal alignment caused by complex backgrounds. To achieve this, we build our method upon the LAE-DINO framework, proposing a DynaGraph-CrossAtt Network. Our DynaGraph-CrossAtt innovatively introduces a DSG convolution and a BiCMA mechanism to realize robust alignment of cross-modal representations through dual-module co-optimization. Fig. 2 illustrates our DynaGraph-CrossAtt overall framework.

Given the training dataset $\mathcal{D}_{base}$, the DVC module in LAE-DINO generates a vocabulary set $\mathcal{V}_{base}$ for each training iteration, which consists of both positive and negative vocabularies. The image I and the vocabulary $\mathcal{V}_{base}$ are, respectively, input into the vision backbone $\mathcal{B}_v$ (i.e., Swin-Transformer [48]) and the text backbone $\mathcal{B}_t$ (i.e., BERT [49]) to obtain the vision features $\boldsymbol{X}_v \in \mathbb{R}^{N_v \times D}$ and text features $\boldsymbol{X}_t \in \mathbb{R}^{N_t \times D}$, where N_v and N_t represent the number of image and text tokens, D indicating the dimension of features. Before the Transformer encoder, the vision and text features are fed into the visual-guided text prompt learning module (VisGT) in LAE-DINO to enhance the representation of text features. However, VisGT is still deficient in explicitly modeling complex semantic associations among text features, and the unidirectional VisGT is susceptible to interference from background noise in vision features. Therefore, on this basis, we introduce the core innovation module, DynaGraph-CrossAtt, which includes DSG and BiCMA, after VisGT to enhance cross-modal interaction. As shown in Fig. 2, the DSG module constructs a semantic graph structure based on a graph convolutional network (GCN), dynamically adjusts the edge weights through the semantic similarity between nodes, realizes the quantitative modeling of semantic relationships, and generates highly discriminative embedding for novel classes. Then, BiCMA receives the enhanced text semantic embeddings and performs cross-modal interaction with vision features, achieving a soft alignment of "semantic-guided localization and vision feedback optimization" through a bidirectional attention mechanism. As the two core modules of DynaGraph-CrossAtt, DSG, and BiCMA assume the roles of semantic enhancement and cross-modal interaction, respectively. The two complement each other and jointly improve the performance of open-vocabulary detection in complex aerial scenarios. Finally, similar to LAE-DINO, the vision and text features augmented by the DynaGraph-CrossAtt module are input into the Transformer encoder–decoder and detection head to obtain object localization and open-vocabulary classification predictions.

B. DSG Graph Convolutional Module

To quantitatively model the complex semantic relationships between classes for enhancing the discriminative and context-aware capabilities of text embedding, especially novel

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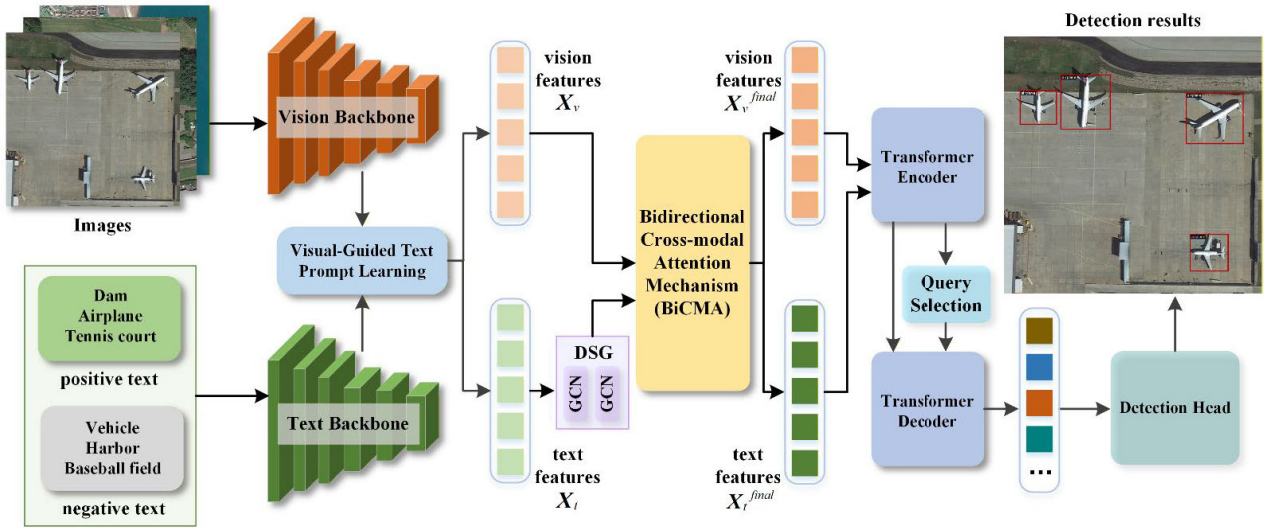


Fig. 2. Framework of our DynaGraph-CrossAtt, which contains a DSG graph convolution and a BiCMA mechanism.

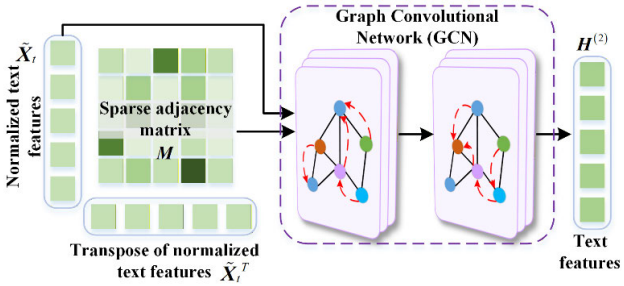


Fig. 3. Structure of the DSG graph convolution module.

classes, we design a DSG graph convolution module. It is important to emphasize that the semantic graph in DSG is constructed entirely dynamically based on the data within each training batch. In DSG, each node represents a class and is initialized by its text embedding. Edges are created dynamically by computing the similarity between all classes and connecting nodes only when the similarity exceeds a predefined threshold. This DSG graph construction allows the graph structure to adapt to different data distributions and class combinations. Once the graph structure is established, we employ a learnable GCN to propagate information over this dynamic graph. The weights of the GCN are optimized during training to learn how to effectively aggregate information from semantically related neighbors, thereby enhancing the representation of each class. Compared with static semantic graph methods [50], [51], [52], [53] that rely on predefined category relationships or external knowledge bases, DSG has three advantages. First, its dynamic graph construction mechanism adapts to changing category combinations in open-vocabulary scenarios. Second, by propagating fine-grained semantic relationships (such as component interactions and spatial dependencies) through graph convolution, it enriches the context-aware ability of text embeddings. Third, it enables knowledge transfer from base classes to novel classes through graph connections, effectively alleviating the problem of insufficient representation for novel classes caused by the lack of visual annotations.

In particular, as shown in Fig. 3, in each training batch, given the text embedding features $\boldsymbol{X}_t \in \mathbb{R}^{N_t \times D}$, the semantic similarity matrix $\boldsymbol{S}$ of the normalized text embedding is computed, i.e.,

$$\tilde{\boldsymbol{X}}_t = \frac{\boldsymbol{X}_t}{\|\boldsymbol{X}_t\|_2}$$
$$\boldsymbol{S} = \tilde{\boldsymbol{X}}_t \tilde{\boldsymbol{X}}_t^T \in \mathbb{R}^{N_t \times N_t}. \tag{1}$$

Then, constructing a sparse adjacency matrix $\boldsymbol{M}$ based on the similarity threshold τ

$$\varepsilon = \left\{(i,j) \,|\boldsymbol{S}_{ij} \geq \tau \wedge i \neq j\right\}$$
$$\boldsymbol{M}_{ij} = \begin{cases} \boldsymbol{S}_{ij}, & if\ (i,j) \in \varepsilon \\ 0, & otherwise \end{cases} \tag{2}$$

where τ is a configurable similarity threshold (default 0.5), which controls the sparsity of the graph, reduces noise interference through sparse connections, and enables the model to adapt to different semantic distribution scenarios. The contextual information of the cross-category nodes is dynamically aggregated based on the semantic similarity, enabling novel class nodes to actively capture the discriminative features of related classes.

After dynamically constructing the semantic graph based on the text embedding similarity, a two-layer GCN is adopted to achieve the dissemination of semantic information, which is formulated as

$$\boldsymbol{H}^{(1)} = \mathrm{ReLU}\left(\boldsymbol{M}\tilde{\boldsymbol{X}}_t \boldsymbol{W}_1\right)$$
$$\boldsymbol{H}^{(2)} = \boldsymbol{M}\boldsymbol{H}^{(1)}\boldsymbol{W}_2 \tag{3}$$

where both $\boldsymbol{W}_1 \in \mathbb{R}^{D \times h}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{h \times h}$ are learnable weights, h is a hidden layer dimension. Through dynamic semantic graph construction and graph convolutional propagation, the output text embedding $\boldsymbol{H}^{(2)}$ consists of context-enhanced semantic representations.

In addition, the quantitative modeling in DSG is reflected in two key aspects. First is the quantifiable relationship strength. The formation of edges is controlled by the quantifiable cosine similarity threshold between node embeddings, converting qualitative semantic affinity into measurable and adjustable

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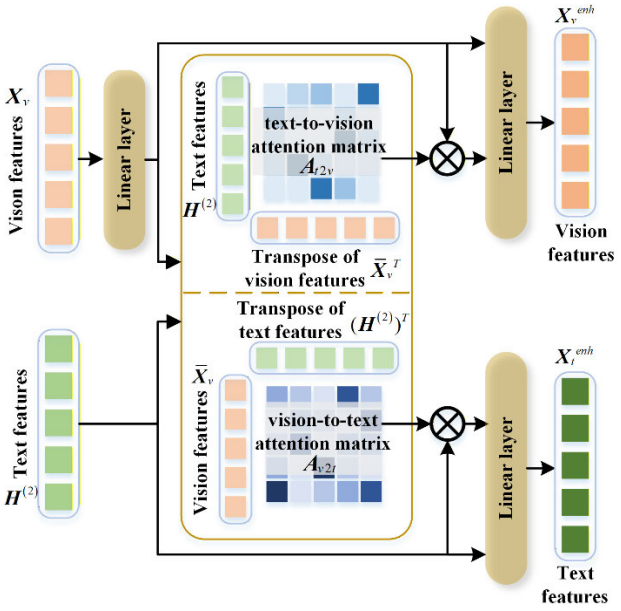


Fig. 4. Structure of the BiCMA mechanism module.

graph topological structures. Second is the learnable information aggregation. The subsequent GCN uses learnable weight matrices to quantitatively weight and aggregate the features of neighboring nodes, where the aggregation weights come from the normalized adjacency matrix (which encodes the quantified similarity). Throughout the entire process, from the creation of edges to the update of nodes, it is based on clear mathematical operations, enabling the module to learn and optimize semantic relationships in a data-driven and quantitative manner, avoiding reliance on external knowledge bases, and allowing the model to flexibly adapt to the requirements of open-vocabulary scenarios.

C. Bidirectional Cross-Modal Attention Mechanism

Aiming at the problem of insufficient robustness of cross-modal semantic alignment in complex backgrounds in aerial images, we propose a BiCMA mechanism, which establishes a dynamic soft-alignment paradigm of "semantic-guided localization and vision feedback optimization" through a dual-branch collaborative architecture of vision-to-text and text-to-vision, as shown in Fig. 4. The "vision feedback optimization" refers to the core process within the vision-to-text branch, which utilizes region vision features as feedback signals. Rather than directly modifying the values of text embeddings, it dynamically optimizes the attention weights of individual text nodes during subsequent fusion. By calculating the correlation between vision features and text nodes, this branch enhances the weights of nodes that are strongly relevant to the current vision content while suppressing irrelevant ones, thereby achieving semantic focusing. Subsequently, "semantic-guided localization" is executed by the text-to-vision branch, which leverages the semantic graph structure to guide the model in focusing on object salient regions and suppressing noise from complex backgrounds.

Concretely, performing vision feature projection can be expressed as

$$\bar{\boldsymbol{X}}_v = \boldsymbol{W}_v \boldsymbol{X}_v \tag{4}$$

where $\boldsymbol{W}_3 \in \mathbb{R}^{D\times h}$ is the learnable weight. For the vision-to-text branch, cross-attention calculation is performed to capture the semantic association between vision and text features and obtain the attention weight matrix $\boldsymbol{A}_{v2t}$

$$\begin{aligned} \boldsymbol{Q}_v &= \bar{\boldsymbol{X}}_v, \quad \boldsymbol{K}_t = \boldsymbol{H}^{(2)}, \boldsymbol{V}_t = \boldsymbol{H}^{(2)} \\ \boldsymbol{A}_{v2t} &= Softmax\left(\frac{\boldsymbol{Q}_v \boldsymbol{K}_t^T}{\sqrt{h}}\right) \in \mathbb{R}^{N_v \times N_t}. \end{aligned} \tag{5}$$

The attention weight matrix $\boldsymbol{A}_{v2t}$ is normalized to a probability distribution by *Softmax*, indicating the intensity of attention at each position of the vision feature to different parts of the text feature. By weighting and summing the text feature $\boldsymbol{V}_t$ with the attention weight matrix $\boldsymbol{A}_{v2t}$, the output fuses the information in the text feature that is most relevant to the current vision feature, thus realizing visually-driven semantic node weight adjustment and enhancing text feature representation

$$\boldsymbol{X}_t^{enh} = \boldsymbol{W}_t \left(\boldsymbol{A}_{v2t}^T \boldsymbol{V}_t + \boldsymbol{H}^{(2)}\right) \tag{6}$$

where $\boldsymbol{W}_t \in \mathbb{R}^{h\times D}$ is the learnable weight.

For the text-to-vision branch, we perform the reverse attention computation to obtain the text-to-vision attention weight matrix $\boldsymbol{A}_{t2v}$, which is formulated as

$$\begin{aligned} \boldsymbol{Q}_t &= \boldsymbol{H}^{(2)}, \quad \boldsymbol{K}_v = \bar{\boldsymbol{X}}_v, \boldsymbol{V}_v = \bar{\boldsymbol{X}}_v \\ \boldsymbol{A}_{t2v} &= Softmax\left(\frac{\boldsymbol{Q}_t \boldsymbol{K}_v^T}{\sqrt{h}}\right) \in \mathbb{R}^{N_t \times N_v}. \end{aligned} \tag{7}$$

By using the attention weight matrix $\boldsymbol{A}_{t2v}$, text semantic guides visual attention focus, with high weights focusing on areas of strong relevance to the task and low weights weakening background or noise interference, which in turn enhances the vision feature representation

$$\boldsymbol{X}_v^{enh} = \boldsymbol{W}_v \left(\boldsymbol{A}_{t2v}^T \boldsymbol{V}_v + \bar{\boldsymbol{X}}_v\right) \tag{8}$$

where $\boldsymbol{W}_v \in \mathbb{R}^{h\times D}$ is the learnable weight.

We use learnable balance parameters to perform weighted fusion of the original features ($\boldsymbol{X}_v$ and $\boldsymbol{X}_t$) and enhanced features ($\boldsymbol{X}_v^{enh}$ and $\boldsymbol{X}_t^{enh}$) as the final output, as shown below

$$\begin{aligned} \boldsymbol{X}_v^{final} &= \alpha \cdot \boldsymbol{X}_v + \beta \cdot \boldsymbol{X}_v^{enh} \\ \boldsymbol{X}_t^{final} &= \alpha_1 \cdot \boldsymbol{X}_t + \beta_1 \cdot \boldsymbol{X}_t^{enh} \end{aligned} \tag{9}$$

where the parameters $\{\alpha, \beta, \alpha_1, \beta_1\}$ are optimized by gradient descent to dynamically adjust the representation weights between the enhanced features ($\boldsymbol{X}_v^{enh}$ and $\boldsymbol{X}_t^{enh}$) and the original features ($\boldsymbol{X}_v$ and $\boldsymbol{X}_t$).

The iterative update process of BiCMA is tightly coupled with DSG. During each update round of the DSG, BiCMA utilizes the vision features generated in the current iteration to optimize the weights of text nodes, and the optimized text representations subsequently participate in the next round of graph convolution updates. This iterative collaborative optimization ensures that cross-modal alignment can gradually converge to a more robust state within complex backgrounds.

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D. Loss Function

Our DynaGraph-CrossAtt mainly focuses on enhancing the representation and alignment of vision and text features through DSG and BiCMA, and does not introduce a new loss function design. Therefore, the overall optimization objective follows the configuration of the baseline LAE-DINO. The total loss $\mathcal{L}_{total}$ consists of classification loss $\mathcal{L}_{cls}$, localization loss $\mathcal{L}_{loc}$, and contrastive loss $\mathcal{L}_{VisGT}$ and is formulated as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{loc} + \lambda_2 \mathcal{L}_{VisGT} \tag{10}$$

where $\mathcal{L}_{cls}$ is the binary cross-entropy loss, which measures the discrepancy between the predicted class and the ground-truth class. $\mathcal{L}_{loc}$ is composed of both L1 loss and GIoU loss, which are used to optimize the position and size of the predicted bounding boxes. $\mathcal{L}_{VisGT}$ supervises image-level semantic alignment learning by pulling the representations of positive pairs closer together while pushing those of negative pairs apart. The hyperparameters λ_1 and λ_2 are used to balance the importance of the three types of losses.

Although the DSG and BiCMA modules do not introduce additional loss terms themselves, they are optimized in an end-to-end manner by participating in the gradient backpropagation of the aforementioned loss functions. In particular, the enhanced text features generated by the DSG module are directly used in computing $\mathcal{L}_{cls}$ and $\mathcal{L}_{VisGT}$. The fused features output by the BiCMA module influence the predictions of both the classification branch and the localization branch. Consequently, the parameters of the two newly added modules are optimized through $\mathcal{L}_{total}$, ensuring that semantic enhancement and cross-modal interaction work collaboratively toward the final open-vocabulary detection task.

IV. Experiments

A. Experimental Setup

To ensure a fair and controlled comparison and to attribute all performance gains explicitly to our architectural innovations, rather than to other factors, all experimental settings in this experimental section, including the input resolution, channel width, Transformer encoder/decoder structure, and backbone network, strictly follow the general configuration of the baseline (i.e., LAE-DINO). This controlled-variable approach is intended to focus the validation on the effectiveness of the proposed modules themselves. In addition, we follow the LAE-DINO to pretrain our method on the LAE-1M dataset and evaluate the detection performance of our method under open-vocabulary scenarios on DIOR [43], DOTAv2.0 [54], and LAE-80C [34] benchmarks. We utilize AP50 and mAP as the evaluation metrics. AP50 represents the average precision at an IoU threshold of 0.5. mAP is a comprehensive performance metric under multiple IoU thresholds (from 0.5 to 0.95). To ensure a fair comparison with the baseline LAE-DINO, we maintain consistency with its settings. On the DIOR dataset, we follow the use of AP50, while on the DOTAv2.0 and LAE-80C datasets, we adopt mAP to more comprehensively evaluate the model's performance in complex remote sensing scenarios. In addition, to ensure the fairness of comparisons, the experimental results of the comparison algorithms are directly cited from the baseline LAE-DINO paper.

LAE-1M contains 1 000 000 high-quality annotated instances, encompassing diverse and typical remote sensing object classes such as aircraft, ships, vehicles, buildings, airports, ports, and farmland. This dataset consists of two complementary subsets, namely LAE-FOD and LAE-COD. Thereinto, LAE-FOD integrates and unifies eight mainstream remote sensing datasets (including DOTA-v2.0, DIOR, FAIR1M [55], NWPU VHR-10 [56], HRSC2016 [57], and so on), ensures category balance and spatial resolution consistency through image cropping, format alignment (unified to COCO format [58]), and a random sampling process. LAE-COD is a coarse-grained dataset constructed by LAE-Label Engine on AID [59], NWPU-RESISC45 [60], SLM [61], and EMS datasets [34].

DIOR comprises 23 463 high-resolution remote sensing images with dimensions of 800×800 pixels, featuring spatial resolutions ranging from 0.5 to 30 m. It encompasses 20 common object classes (e.g., aircraft, airport, baseball diamond, and bridge) with a total of 192 472 annotated target instances, characterized by multisource heterogeneity, object diversity, and interclass similarity.

DOTAv2.0 contains 11 268 aerial images covering 18 common classes (e.g., aircraft, ships, tanks, vehicles, and so on), totaling 1 793 658 labeled instances. Compared to the initial version, v2.0 significantly expands both the number of images and category diversity.

LAE-80C is an emerging benchmark dataset for the open-vocabulary target detection task in remote sensing. It comprises 80 fine-grained classes and supports model generalization to detect novel classes absent from the training set during testing, thereby emulating real-world requirements for "arbitrary object detection."

1) Implementation Details: To ensure the comparability of the experimental results, we strictly follow the experimental configuration of the baseline for model training and optimization. In particular, the model uses a stochastic gradient descent (SGD) optimizer with an initial learning rate set to 0.02, a weight decay coefficient of 0.0001, and a momentum parameter of 0.9. The pretraining experiments are executed on two NVIDIA RTX 4090-D GPUs, and the training epochs are 16, and the batch size is set to two for each GPU.

B. Main Results

1) Open-Set Detection: As shown in Table I, our DynaGraph-CrossAtt demonstrates significant performance advantages on the DIOR, DOTAv2.0, and LAE-80C datasets, which validates the generalization performance and fine-grained recognition ability of DynaGraph-CrossAtt in multiple scenarios. In particular, on the DIOR dataset covering 20 object classes, DynaGraph-CrossAtt refreshes the baseline (i.e., LAE-DINO) with 89.6% detection accuracy and achieves a performance leap of 4.1%, demonstrating the stronger generalization ability of DynaGraph-CrossAtt in the semantic space. On the DOTAv2.0 dataset, DynaGraph-CrossAtt also performs outstandingly, and achieves a detection performance of 51.5%, which is 4.7% and 5.5% better than the baseline

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TABLE I

OPEN-SET DETECTION RESULTS EVALUATED ON DIOR, DOTAV2.0, AND LAE-80 C BENCHMARKS, WHERE SOME MODELS ARE PRETRAINED ON NATURAL SCENE DATASETS: O365, GOLDG, CC3M, SBU, AND CAP4M

Method	Pre-Training Data	DIOR (AP_{50})	DOTAv2.0 (mAP)	LAE-80C (mAP)
GLIP [35]	O365,GoldG,CC3M,SBU	1.1	0.2	0.1
GLIP with DVC [35]	LAE-1M	82.8	43.0	16.5
GroundingDINO [14]	O365,GoldG,Cap4M	0.3	0.3	0.1
GroundingDINO with DVC [14]	LAE-1M	83.6	46.0	17.7
LAE-DINO [34]	LAE-1M	85.5	46.8	20.2
DynaGraph-CrossAtt (ours)	LAE-1M	89.6	51.5	22.7

and GroundingDINO, respectively, and fully demonstrates its scene adaptability and object localization robustness under complex aerial perspectives. On the LAE-80C dataset designed for open-vocabulary detection, our DynaGraph-CrossAtt significantly leads with a detection accuracy of 22.7% and achieves an improvement of 2.5% and 5% compared to the baseline and GroundingDINO, respectively, demonstrating excellent fine-grained discrimination capabilities. The quantitative results confirm that the proposed method significantly improves the comprehension and detection capacity of the detector for unseen classes and complex scenes by building a DSG graph convolution and a BiCMA. In addition, to visually demonstrate the specific improvements brought by the proposed core modules (DSG and BiCMA), we conduct visualization analysis, focusing on comparisons with the baseline method LAE-DINO. As shown in Fig. 5, our DynaGraph-CrossAtt consistently demonstrates superior performance in detection accuracy, robustness, and fine-grained recognition across various challenging remote sensing scenarios. In particular, the results of LAE-DINO reveal its limitations in certain complex situations, such as localization drift for distant vehicles, missed detections of bridges, imprecise detection of objects like tennis courts in dense building areas, and insufficient perception of small-scale objects (e.g., swimming pools) in residential areas. On the contrary, our DynaGraph-CrossAtt shows significant improvement in all these scenarios. For example, DynaGraph-CrossAtt successfully detects the tennis courts under tree shadows (e.g., fourth row and second column), whereas the baseline method fails in this case. This validates the effectiveness of the text-to-vision branch in BiCMA, which utilizes the semantic graph to guide visual attention, thereby suppressing background noise like shadows. Our method also excels at distinguishing confusing categories. For instance, it accurately discriminates between bridges and overpasses in the first column of the fourth row, demonstrating robustness against interclass similarities. In addition, the baseline falsely detects mirror-like building facades as vehicles (third row and third column), while our DynaGraph-CrossAtt avoids this error. This benefits from the vision-to-text branch, which dynamically adjusts weights to suppress semantic nodes irrelevant to the current visual content, thereby preventing semantic drift caused by background texture similarities. In dense and complex scenes, such as bridges in dense building areas, our method reduces missed detections and improves localization accuracy, reflecting the capability of the semantic-guided localization mechanism to help the model focus on object regions and resist interference from cluttered background information in complex spatial layouts. In summary, the qualitative comparison in Fig. 5 corroborates the quantitative results, not only verifying the effectiveness of our DynaGraph-CrossAtt in cross-modal semantic alignment and spatial relationship modeling but also demonstrating the superiority of our method in handling complex remote sensing detection tasks, particularly in its ability to counteract background interference from multiple perspectives.

2) Closed-Set Detection: Aligning with the baseline methods, we also conduct fine-tuning experiments and perform closed-set detection performance comparisons with various general object detection methods on both the DIOR and DOTA v2.0 datasets. As illustrated in Table II, our DynaGraph-CrossAtt exhibits distinct performance characteristics across different datasets. On the relatively standardized DIOR dataset, our DynaGraph-CrossAtt achieves a detection accuracy of 91.2%, which is slightly lower than that of the baseline by about 1%. This is primarily attributed to the lower complexity of the DIOR dataset, where high-performance models have approached the performance ceiling, making it difficult to achieve significant gains. Nevertheless, this result still remains at an exceptionally high-performance level (>90%). Conversely, on the DOTA v2.0 test set, which better reflects the complexity of real-world scenarios, our method demonstrates superior performance, reaching 59.3%. This significantly outperforms the baseline method (57.9%), GroundingDINO (55.1%), and GLIP-FT (50.6%). This improvement is attributed to the DSG in our DynaGraph-CrossAtt, which effectively models spatial dependencies among dense and interleaved objects, and the BiCMA module, which better utilizes text semantics to distinguish object classes within complex backgrounds. In summary, these results indicate that our DynaGraph-CrossAtt possesses a distinct advantage in handling highly challenging scenarios while maintaining an exceptionally high baseline performance. This not only enhances open-vocabulary detection capabilities but also validates its effectiveness in closed-set detection tasks, particularly in complex scenarios, thereby achieving a superior balance between open vocabulary and closed-set performance. In addition, it should be noted that the absence of results for some algorithms on the DOTAv2.0 dataset, as illustrated in Table II, is because the original papers of these algorithms have not published their evaluation results or lacked

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Fig. 5. Qualitative comparison between our DynaGraph-CrossAtt and baseline (i.e., LAE-DINO).

reproducible open-source code/models on the DOTAv2.0 dataset. Therefore, for the sake of rigor, we choose to directly quote the existing results in the original paper and do not present the results that cannot be reproduced. This operation is also consistent with LAE-DINO.

C. Ablation Studies

To validate the effectiveness of our DynaGraph-CrossAtt, we carry out ablation experiments on DIOR, DOTAv2.0, and LAE-80C datasets (see Table III), where T2V denotes the text-to-vision branch in BiCMA and V2T denotes the vision-to-text branch in BiCMA. First, the introduction of the DSG module alone on top of the baseline achieves 2.8%, 5.5%, and 2% performance improvement on the DIOR, DOTAv2.0, and LAE-80C datasets, respectively, which verifies that the text semantic graph enhances the capability of modeling interclass semantic associations through dynamic node updating as well as graph structure propagation mechanisms. We choose to use the with/without DSG setup for comparison to validate the effectiveness of the dynamic graph at the most fundamental level, thereby avoiding the introduction of additional confounding variables that could arise from incorporating other complex comparative methods. The performance gains demonstrated by DSG on the challenging datasets provide empirical support for the clear advantages of its design over the baseline LAE-DINO of unstructured semantic modeling. Second, the introduction of the BiCMA mechanism alone also brought considerable performance improvements, 3%, 5.9%, and 2.1% on DIOR, DOTAv2.0, and LAE-80C, respectively. The significant performance improvement brought by BiCMA, particularly on datasets with complex backgrounds such as DOTAv2.0, validates the effectiveness of its design. Through the bidirectional attention mechanism, BiCMA alleviates issues of feature confusion and semantic misalignment caused by complex backgrounds, thereby providing crucial support for the model in handling background interference in open-vocabulary scenarios. Furthermore, we conduct combination experiments of DSG with two subbranches of BiCMA (T2V or V2T), respectively, and the results show that these

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TABLE II

EVALUATIONS ON DIOR AND DOTAV2.0 TEST SETS UNDER CLOSED-SET DETECTION SETTINGS. SOME MODELS ARE PRETRAINED ON REMOTE SENSING DATASETS SUCH AS GEOIMAGENET, SENTINEL-2, TOV-NI, TOV-R, FMOW, SATLASPRETRAIN, MILLIONAID, RINGMOPRETRAIN, AND MULTIMODAL RSI

Method	Backbone	Pre-Training Data	Fine-Tuning	
			DIOR (AP_{50})	DOTAv2.0 (mAP)
Generic Object Detection				
GASSL [62]	ResNet-50	-	67.40	-
CACO [63]	ResNet-50	Sentinel-2	66.91	-
TOV [64]	ResNet-50	TOV-NI,TOV-R	70.16	-
Scale-MAE [65]	ViT-L	FMoW	73.81	-
SatLas [66]	Swin-B	SatlasPretrain	74.10	-
RingMo [67]	Swin-B	RingMoPretrain	75.90	-
SkySense [68]	Swin-H	multi-modal RSI	78.73	-
MTP [69]	Swin-H	MillionAID	81.10	-
Open-Vocabulary Object Detection				
GLIP-FT [35]	Swin-T	O365,GoldG,CC3M,SBU	87.8	50.6
GroudingDINO-FT [14]	Swin-T	O365,GoldG,Cap4M	90.4	54.0
GroudingDINO-FT [14]	Swin-T	LAE-1M	91.1	55.1
LAE-DINO-FT [34]	Swin-T	O365,GoldG,Cap4M	92.0	55.5
LAE-DINO-FT [34]	Swin-T	LAE-1M	92.2	57.9
DynaGraph-CrossAtt (Ours)	Swin-T	LAE-1M	91.2	59.3

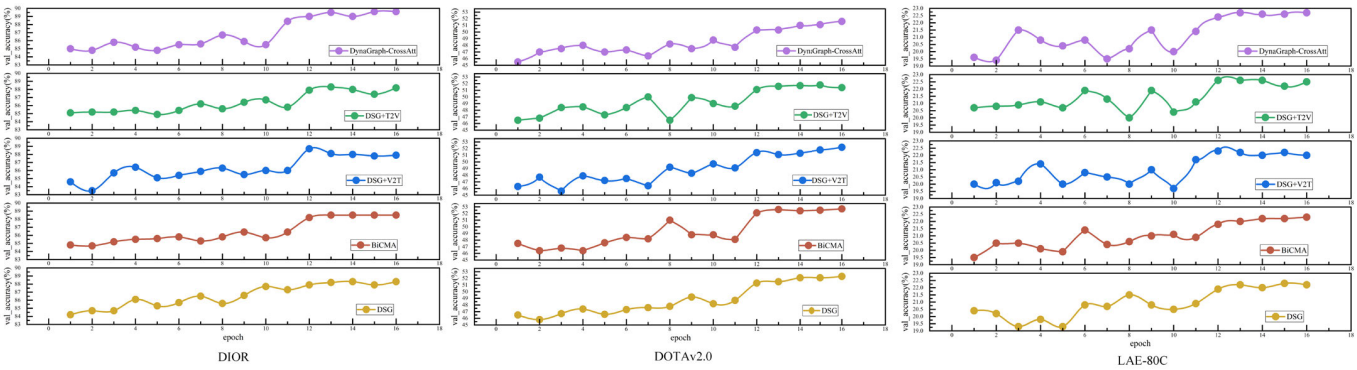


Fig. 6. Validation accuracy evolution of each ablated variant over successive training epochs.

TABLE III

ABLATION RESULTS ON THE DIOR, DOTAV2.0, AND LAE-80 C UNDER OPEN-SET DETECTION SETTINGS

Method	Pre-Training Data	Open-Set Detection		
		DIOR (AP_{50})	DOTAv2.0 (mAP)	LAE-80C (mAP)
Baseline	LAE-1M	85.5	46.8	20.2
+DSG	LAE-1M	88.3	52.3	22.2
+BiCMA	LAE-1M	88.5	52.7	22.3
+DSG+V2T	LAE-1M	87.8	52.2	22.0
+DSG+T2V	LAE-1M	88.2	51.4	22.5
DynaGraph-CrossAtt	LAE-1M	89.6	51.5	22.7

combination schemes deliver consistent performance gains over the baseline (ranging from 1.8% to 5.4%). Finally, the joint deployment of the DSG and BiCMA modules (i.e., our DynaGraph-CrossAtt) achieves superior performance on the DIOR and LAE-80C datasets, 89.6% and 22.7%, respectively, and achieves a competitive performance of 51.5% on the DOTAv2.0 dataset. These results provide a multidimensional validation of our module design. The significant and consistent gains from using DSG or BiCMA alone confirm that both are core components with independent utility. The fact that any "DSG + unidirectional branch" scheme stably outperforms the baseline preliminarily verifies the effectiveness of each attention branch. Significantly, the peak performance achieved by the full model (i.e., BiCMA) on DIOR and LAE-80C strongly supports the design motivation of our BiCMA, which is to achieve the robust cross-modal alignment through a bidirectional closed loop of "vision feedback optimization" (V2T) and "semantic-guided localization" (T2V). In addition, it is noteworthy that on the DOTAv2.0 dataset, we observe

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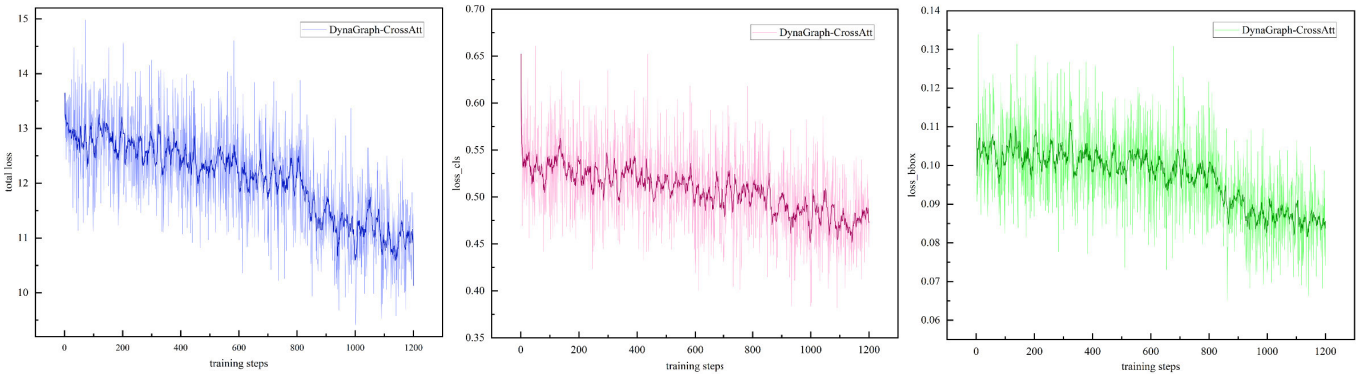


Fig. 7. Convergence curve of the loss function during the pretraining stage of DynaGraph-CrossAtt.

that while the individual use of both the DSG and BiCMA modules brought significant improvements, their integrated performance (51.5%) is slightly lower than the peak performance achieved by either module alone. This does not indicate that the modules are ineffective, but rather reveals an inherent tension between two powerful modules when jointly optimized in extremely complex scenarios. In particular, DSG strives to enhance the discriminability and context-aware capability of novel class text embeddings through graph propagation learning, while BiCMA focuses on establishing precise visual-semantic alignment. Both have strong optimization objectives, and on datasets such as DOTAv2.0, which contain densely arranged small objects and complex scene layouts, their optimization paths may compete, causing the model to converge to a suboptimal equilibrium. In fact, this finding provides strong evidence that both DSG and BiCMA are powerful modules capable of significantly guiding the optimization direction. Despite this subtle competition, our full model (DynaGraph-CrossAtt) still substantially outperforms the baseline (+ 5.7%) on this dataset, fully demonstrating the effectiveness and robustness of the overall framework. Furthermore, the optimal performance of the method on DIOR and LAE-80C, as well as the significant enhancement and competitive results on DOTAv2.0, together fully validate its overall effectiveness, generalizability, and robustness. This finding highlights the challenges of multimodule joint optimization in extremely complex scenarios, pointing toward future research directions, such as designing more sophisticated joint optimization strategies or adaptive module fusion mechanisms to better cope with extreme challenges.

In addition, in order to deeply understand the dynamic properties of different module combinations during the training process, we show the validation accuracy change curves of each ablation combination under different training epochs. As shown in Fig. 6, compared to other combinations and the baseline, our DynaGraph-CrossAtt not only has the highest final accuracy but also its validation accuracy significantly outperforms the other combinations in the middle of the training period and always maintains a monotonous upward trend. The trend of this curve indicates that the dynamic semantic modeling of DSG and the cross-modal alignment capability of BiCMA form an effective synergy during the training process, jointly accelerating the generalization ability formation of the model while maintaining a stable optimization direction throughout the entire training cycle, further validating the rationality of the joint module design.

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY

Method	Parameters	FLOPs
GLIP [35]	232M	434G
Grounding DINO [14]	172M	412G
LAE-DINO [34]	180M	415G
DynaGraph-CrossAtt (ours)	189M	423G

Meanwhile, Fig. 7 demonstrates the loss convergence of our DynaGraph-CrossAtt in the pretraining phase. As shown, the model loss exhibits a fast and smooth downward trend and reaches a steady state in relatively few epochs. This strongly proves that the integrated architecture has advantageous optimization characteristics and training efficiency, and lays a solid foundation for achieving excellent performance in the subsequent fine-tuning phase.

Regarding the analysis of model computational complexity, as illustrated in Table IV, the total number of parameters of our DynaGraph-CrossAtt is approximately 189 M, and the forward propagation FLOPs amount to approximately 423G. Among these, the DSG module, which consists solely of lightweight graph convolutions related to the number of categories, with a small amount of parameters, and because some semantic graph information can be cached during inference, incurs no additional FLOPs. The cross-attention calculation in the BiCMA module, although increasing a modest amount of FLOPs slightly, improves the feature quality, thereby effectively reducing the number of redundant candidate boxes that must be processed in the NMS postprocessing stage, and further enhancing the overall inference efficiency. Compared with the baseline LAE-DINO, our method increases the parameter count by only about 9 M and FLOPs by approximately 8 G. In summary, the proposed DSG and BiCMA modules achieve significant improvements in feature interaction and alignment efficiency with minimal computational overhead, enabling the model to strike a favorable balance between computational complexity and detection accuracy.

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V. Conclusion

In this article, we propose a dynamic semantic DynaGraph-CrossAtt network, which realizes quantitative modeling of fine-grained semantic relationships among text nodes by constructing a text semantic graph and dynamically aggregating node relationships through adaptive GCN, providing highly discriminative embedding of new classes; and designing a BiCMA mechanism, where the vision-to-text branch adjusts the embedding weights of the text by using the vision features to suppress semantic drift, and the text-to-vision branch guides the vision attention through the semantic graph to achieve background noise filtering, forming a collaborative optimization framework of "semantic-guided localization and vision feedback optimization." The effectiveness of DynaGraph-CrossAtt is verified through extensive experiments. While our DynaGraph-CrossAtt achieves significant performance improvements, it still has certain limitations. In particular, in terms of hyperparameter sensitivity, to focus on the core architecture verification, the current experiments follow the mainstream settings in the field. Systematically evaluating the impact of hyperparameters such as different backbone networks, input resolutions, and Transformer variants is crucial for a comprehensive understanding of model robustness and deployment adaptability. We have designated this as a key focus for future research, aiming to further investigate the performance boundaries and generalization patterns of the model under different configurations.

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